

ACCT102 Management Accounting

Week 6

Process costing

Quick Check 6.1

Process costing is used for products that are:

- a. Different and produced continuously.
- b. Similar and produced continuously.
- c. Individual units produced to customer specifications.
- d. Purchased from vendors.

Processing Department

- Any unit in an organization where materials, labor, or overhead are added to the product.
- The activities performed in a processing department are *performed uniformly* on all units of production. Furthermore, the output of a processing department must be *homogeneous*.
- Products in a process costing environment typically flow in a sequence from one department to another.

Two methods in Process Costing computations

1. **Weighted-Average Method**

- calculates unit costs by combining costs and outputs from the current and prior periods

2. **First-In, First-Out Method – FIFO is not examinable in ACCT 102**

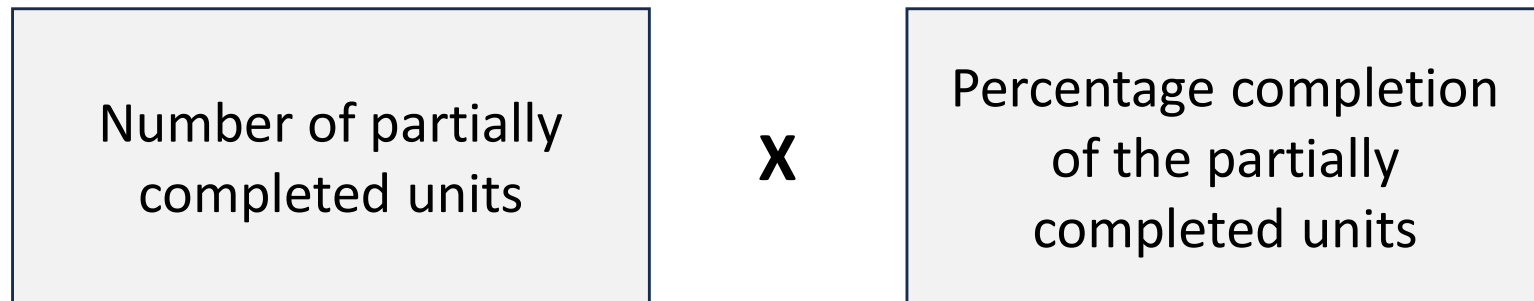
- calculates unit costs based solely on the costs and outputs from the current period

Learning Objective

Understand and calculate
equivalent units

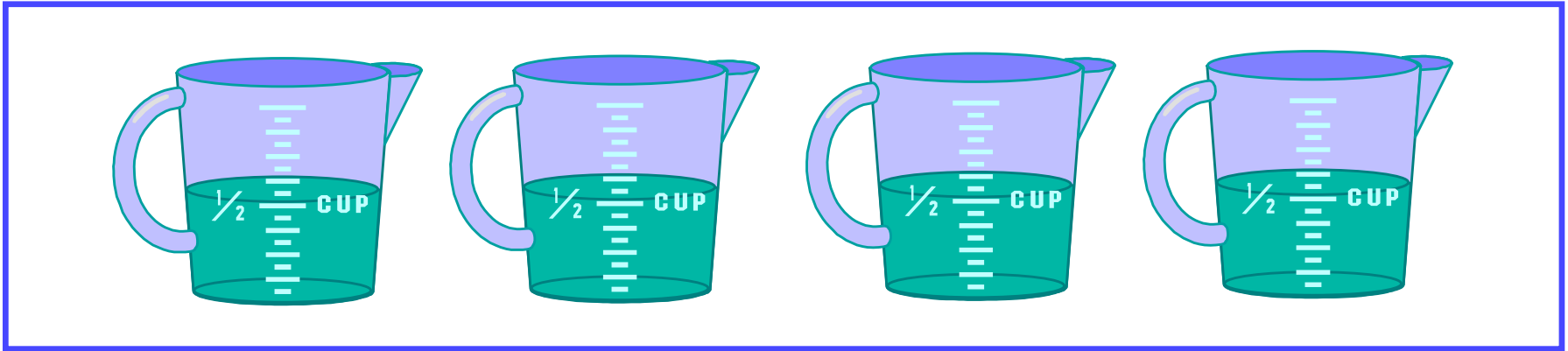
Equivalent Units

Equivalent units are the product of the number of partially completed units and the percentage completion of those units.



Equivalent units are required because a department usually has some partially completed units in its beginning and ending inventory. These partially completed units complicate the determination of a **department's output for a given period** and the unit cost that should be assigned to that output.

Equivalent Units – Illustrative Example



Number of partially
completed units:

4

X

Percentage completion
of the partially
completed units:

50%

Quick Check 6.2

How many equivalent units is there for 10,000 units which are only 70% completed?

- a. 10,000
- b. 70
- c. 7,000
- d. 3,500

Learning Objective

Step 1:
Compute the equivalent units of
production using the weighted-
average method.

Equivalent Units of Production - Weighted-Average

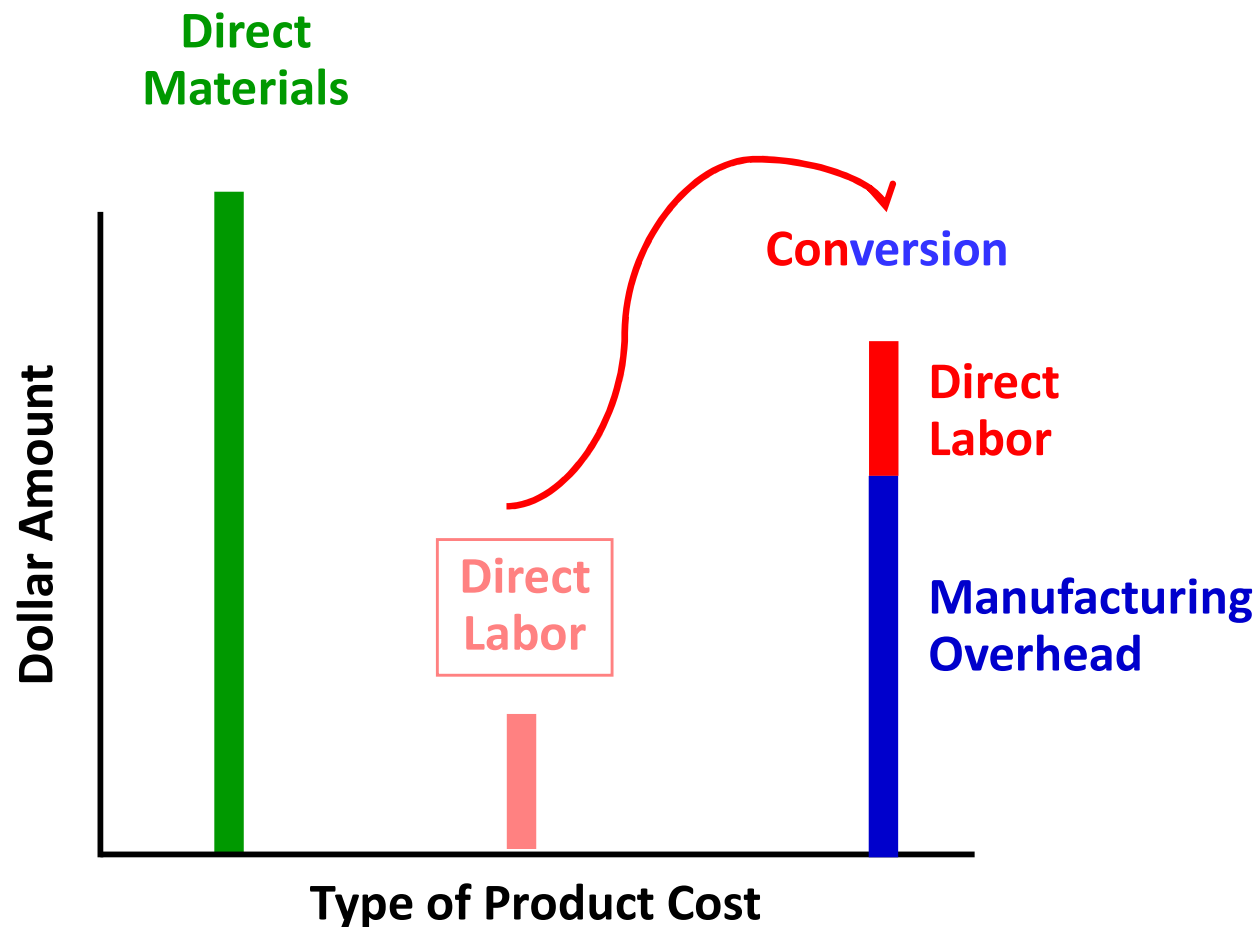
- The weighted-average method makes **no distinction between work done in prior or current periods**
- It blends together the units and costs from prior and current periods.
- Equivalent units of production for a department = **number of units transferred out** (to the next department or finished goods) + the equivalent units in **ending Work in Process Inventory**.

Equivalent units of production *always* equals:
Units completed and transferred
+ Equivalent units remaining in work in process

Treatment of Direct Labor

Direct labor costs are becoming small in comparison to other product costs in process cost systems (due to automation).

Combined with manufacturing overhead and considered as one - **Conversion costs**.



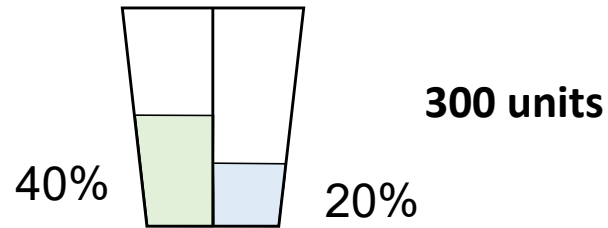
Step 1: Compute the Equivalent Units of Production

Smith Company reported the following activity in Department A for the month of June.

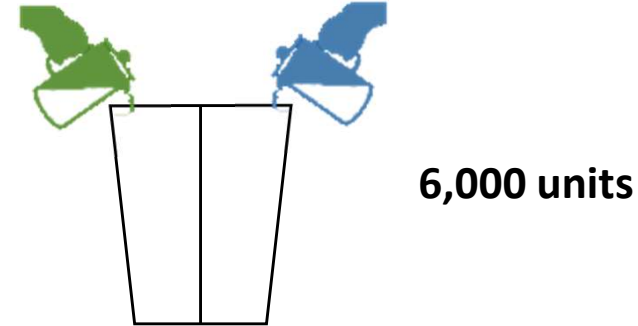
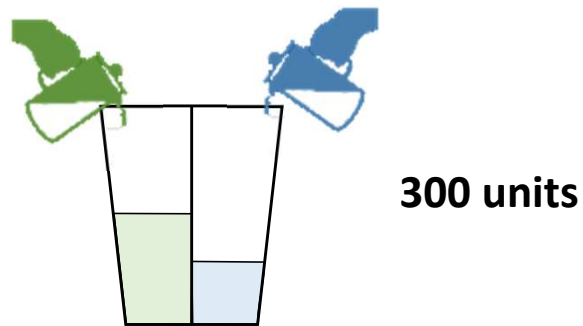
	Units	Percent Completed	
		Materials	Conversion
Work in process, June 1	300	40%	20%
Units started into production in June	6,000		
Units completed and transferred out of Department A during June	5,400		
Work in process, June 30	900	60%	30%

Step 1: Compute the Equivalent Units of Production

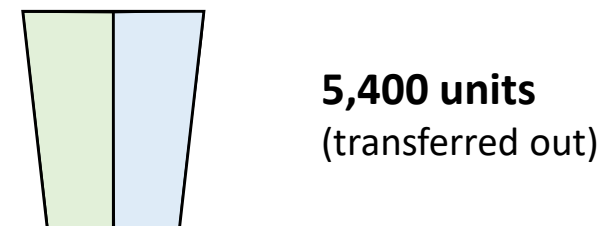
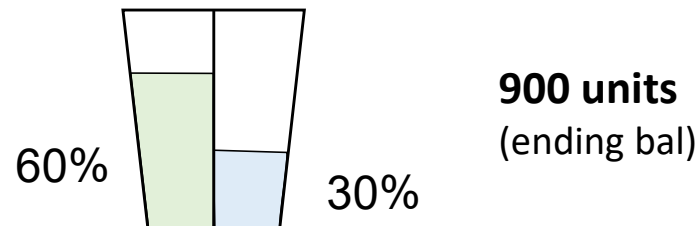
START



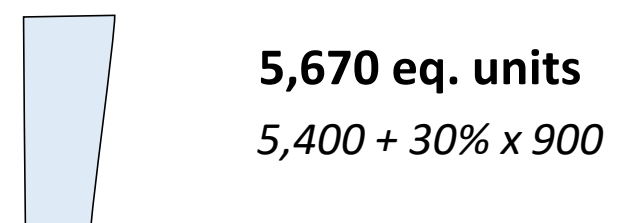
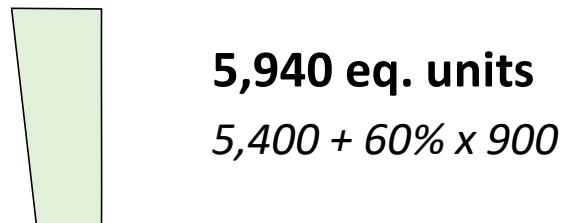
DURING



END



**EQUIVALENT
UNITS OF PDTN**



Step 1: Compute the Equivalent Units of Production

- Calculate the equivalent units completed and transferred out of the Department A in June (5,400 units).
- Identify equivalent units of production in ending WIP with respect to materials for the month (540 units)
- Identify equivalent units of production in ending WIP with respect to conversion for the month (270 units)
- Add both equivalent units of production (materials and conversion) to 5,400 units.

Equivalent units of production *always* equals:
 Units completed and transferred
 + Equivalent units remaining in work in process

	<u>Materials</u>	<u>Conversion</u>
① Units completed and transferred out of the Department A in June	5,400	5,400
Work in process, June 30:		
② 900 units × 60%	540	
③ 900 units × 30%		270
④ Equivalent units of Production in the Department A during June	5,940	5,670

Step 1: Compute the Equivalent Units of Production

MATERIALS

6,000 Units Started

Beginning Work in Process 300 Units 40% Complete	5,100 Units Started and Completed	Ending Work in Process 900 Units 60% Complete
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5,400 Units Completed

$$900 \times 60\%$$

540 Equivalent Units
5,940 Equivalent units of production

Step 1: Compute the Equivalent Units of Production

CONVERSION

6,000 Units Started

Beginning Work in Process 300 Units 20% Complete	5,100 Units Started and Completed	Ending Work in Process 900 Units 30% Complete
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5,400 Units Completed

$$900 \times 30\%$$

270 Equivalent Units
5,670 Equivalent units of production

Quick Check 6.3

At the beginning, Jones had 2,000 units in process, 40% completed.

During the period, Jones started work on 15,000 units. He completed 16,000 units and finished the period with 1,000 units still in process at 30% completion.

How many equivalent units of production did Jones have for the period?

- a. 15,800
- b. 16,000
- c. 16,300
- d. 15,300

Learning Objective

Step 2:
Compute the cost per
equivalent unit using the
weighted-average method.

Step 2: Compute the Cost per Equivalent Unit

Beginning Work in Process Inventory: 300 units

Materials: 40% complete \$ 6,119

Conversion: 20% complete \$ 3,920

Production started during June: 6,000 units

Production completed during June: 5,400 units

Costs added to production in June

Materials cost \$ 118,621

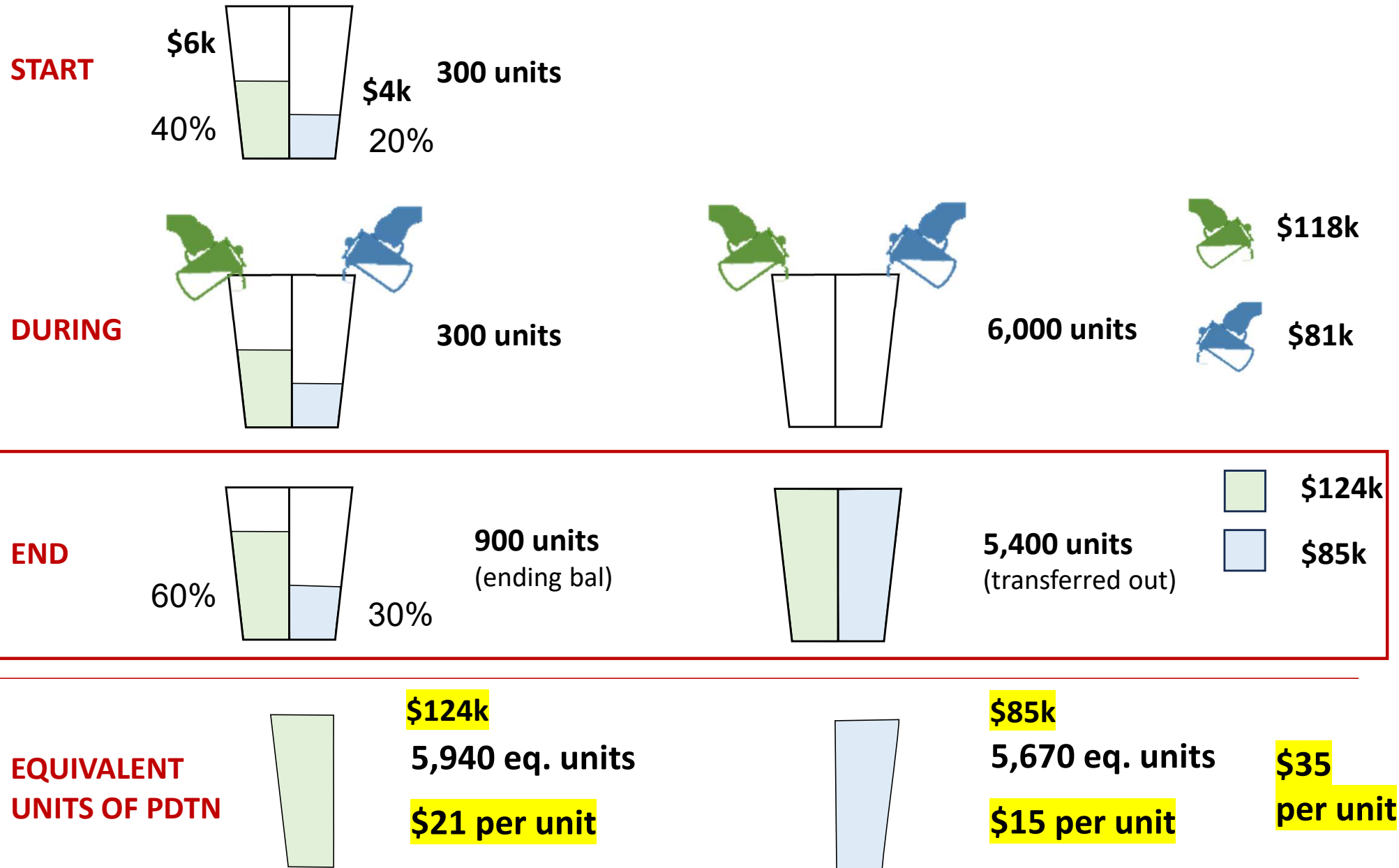
Conversion cost \$ 81,130

Ending Work in Process Inventory: 900 units

Materials: 60% complete

Conversion: 30% complete

Step 2: Compute the Cost per Equivalent Unit



Step 2: Compute the Cost per Equivalent Unit

A separate calculation is made for each cost category in each processing department.

$$\text{Cost per equivalent unit} = \frac{\text{Cost of beginning WIP Inventory} + \text{Cost added during the period}}{\text{Equivalent units of production}}$$

	<u>Total Cost</u>	<u>Materials</u>	<u>Conversion</u>
Cost to be accounted for:			
Work in process, June 1	\$ 10,039	\$ 6,119	\$ 3,920
Cost added in Department A	199,751	118,621	81,130
Total cost	<u><u>\$ 209,790</u></u>	<u><u>\$ 124,740</u></u>	<u><u>\$ 85,050</u></u>
Equivalent units		5,940	5,670
Cost per equivalent unit		<u>\$ 21.00</u>	<u>\$ 15.00</u>

$$\text{Cost per equivalent unit} = \$21.00 + \$15.00 = \$36.00$$

Quick Check 6.4

	<u>Materials</u>
Beg. WIP inventory - equivalent units	100
Units completed and transferred out	800
End WIP inventory - equivalent units	200
Beg. WIP cost	\$500
Cost added	\$2,500

Using the weightage-average method, what is the material cost per equivalent unit for the period?

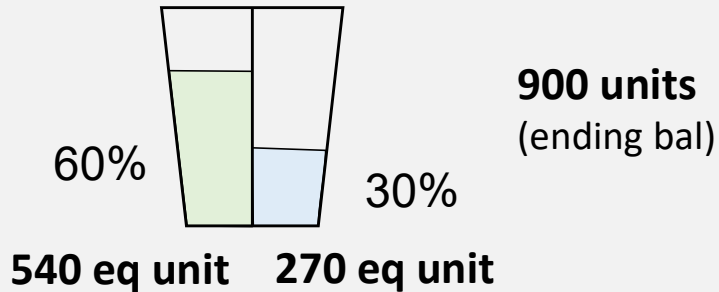
- a. \$3.125
- b. \$3
- c. \$5
- d. \$2.727

Learning Objective

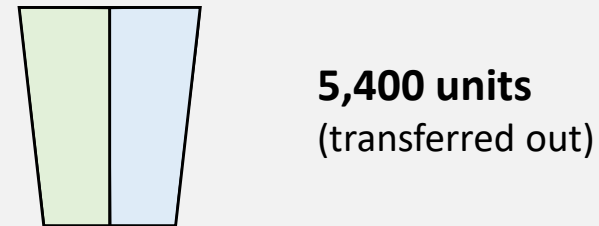
Step 3:
Assign costs to units using the
weighted-average method.

Step 3: Assign Costs to Units

ENDING WIP



UNITS COMPLETED



Department A			
Cost of Ending WIP Inventory and Units Transferred Out			
	Materials	Conversion	Total
Ending WIP inventory:			
Equivalent units	540	270	
Cost per equivalent unit	\$ 21.00	\$ 15.00	
Cost of Ending WIP inventory	\$ 11,340	\$ 4,050	\$ 15,390
Units completed and transferred out:			
Units transferred	5,400	5,400	
Cost per equivalent unit	\$ 21.00	\$ 15.00	
Cost of units transferred out	\$ 113,400	\$ 81,000	\$ 194,400

Learning Objective

Step 4:

Prepare a cost reconciliation report using the weighted-average method.

Step 4: Prepare a Cost Reconciliation Report

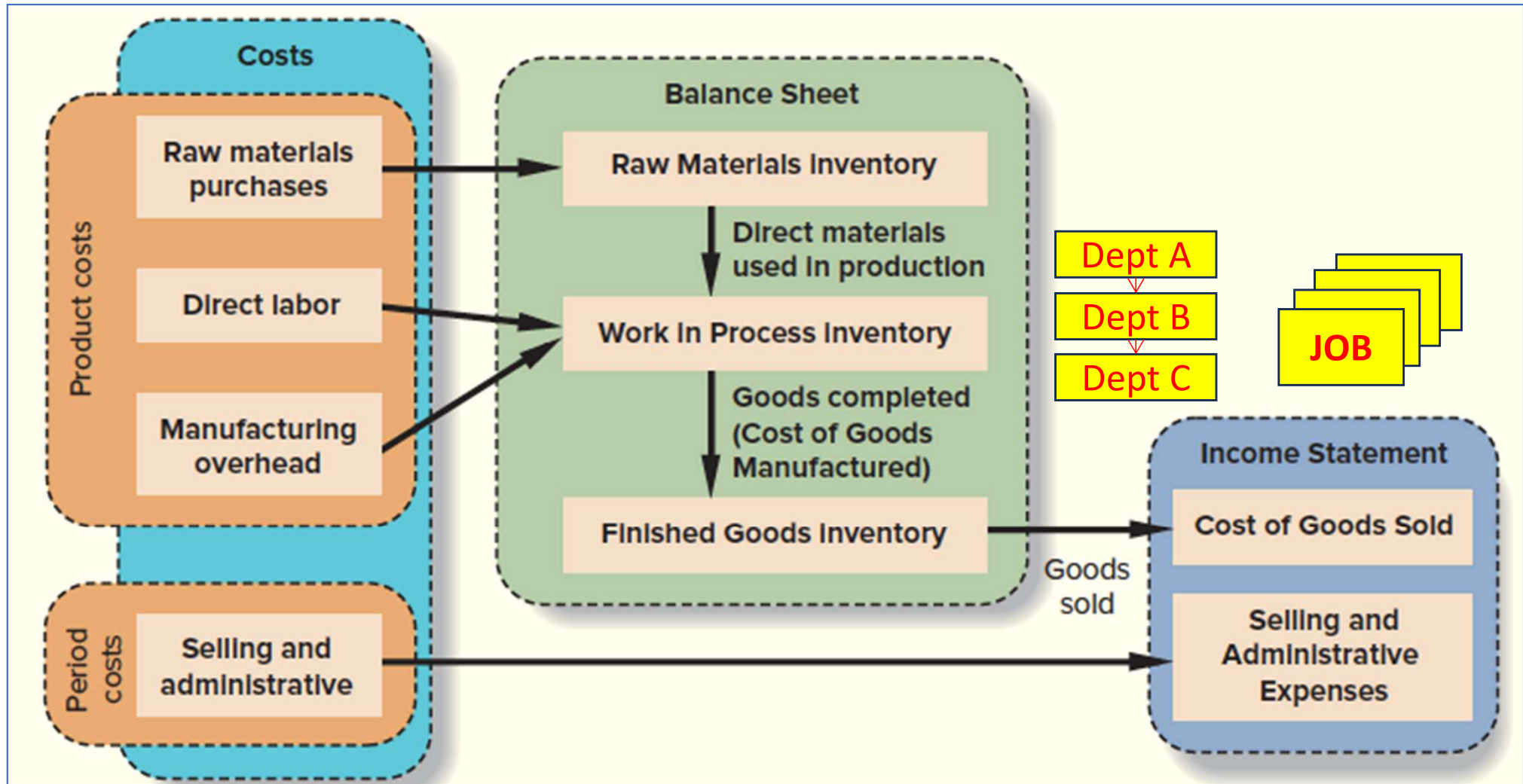
Department A Cost Reconciliation		
Costs to be accounted for:		
Cost of beginning Work in Process Inventory	\$	10,039
Costs added to production during the period		199,751
Total cost to be accounted for	\$	209,790
Cost accounted for as follows:		
Cost of ending Work in Process Inventory	\$	15,390
Cost of units transferred out		194,400
Total cost accounted for	\$	209,790



Operation Costing

- Operation costing is a **hybrid** of job-order and process costing because it possesses attributes of both approaches.
- Used when products have some common characteristics and some individual characteristics.
- Batches of many different products pass through the same processing departments
- Each batch charge for its own materials (similar to job-order costing).
- Labor and overhead cost accumulated by department and assigned to units (similar to process costing).

Process Costing and Job-Order Costing



End of topic